

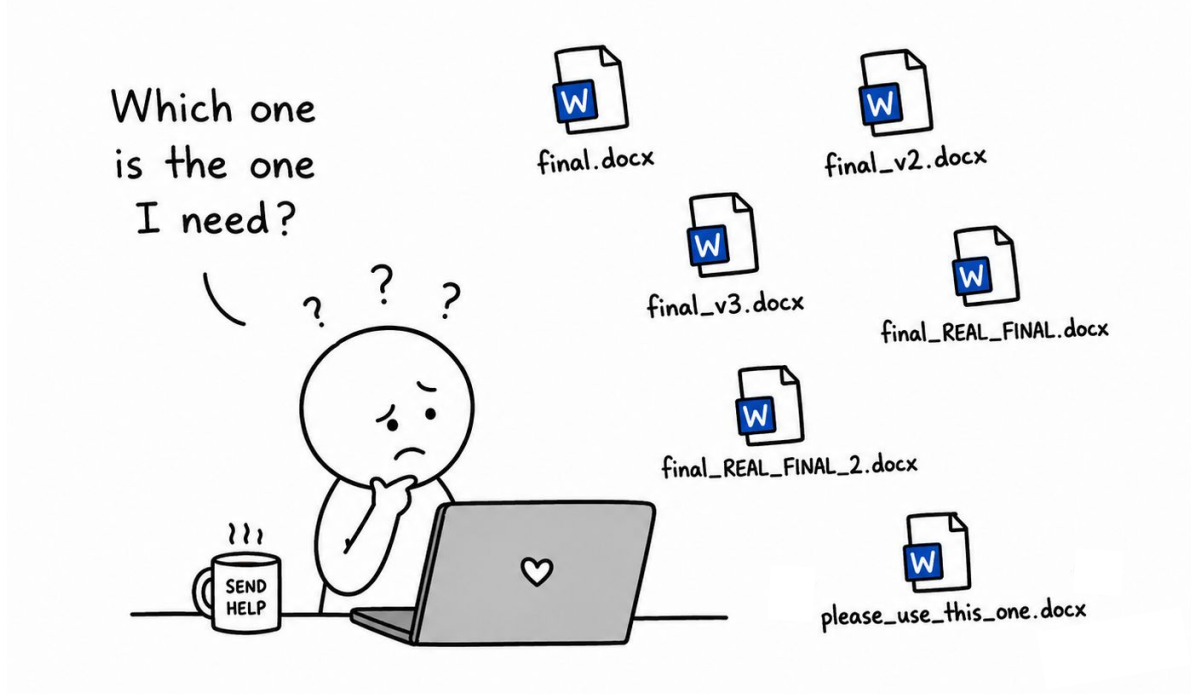


**Play  
with**

**Git**

Wanyi Tang  
USC LA's BeST Program  
06/16/26

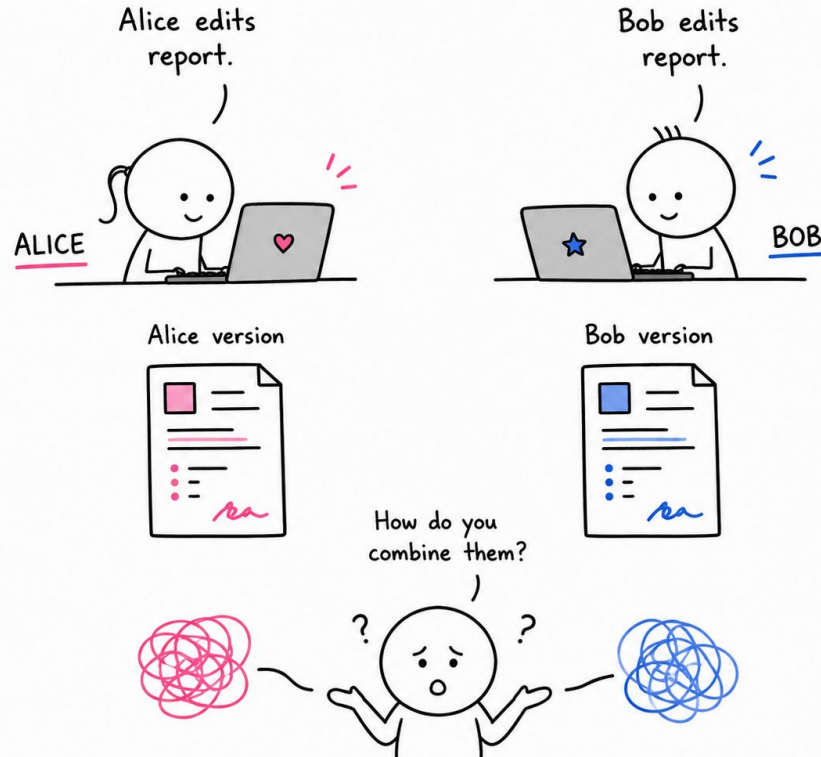
# Scenario 1: Which version?



# Scenario 2: How to go back?



# Scenario 3: How to collab

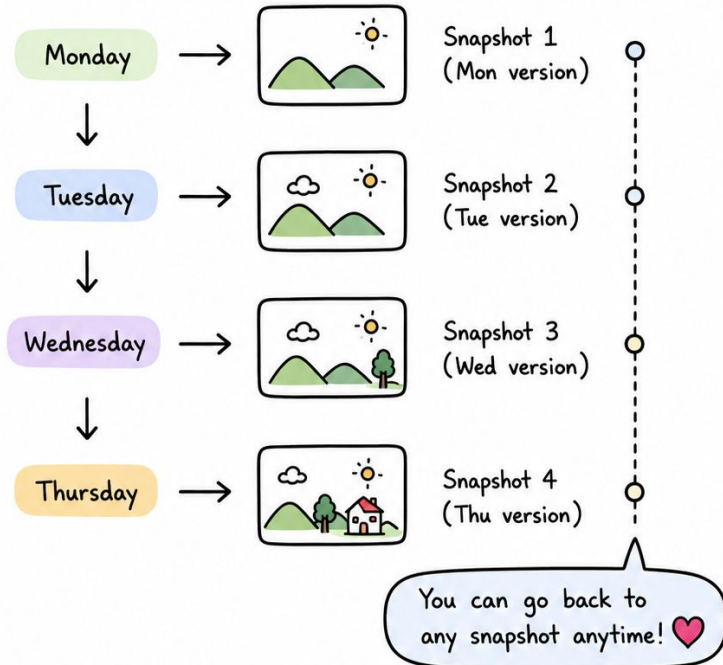
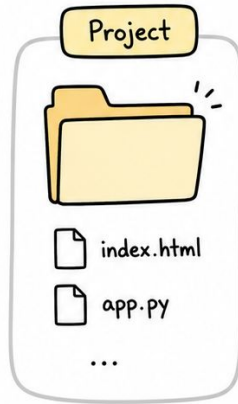
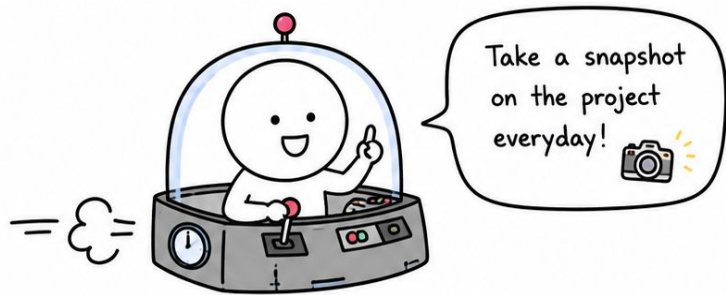


**Git can help with all these scenarios!**



**git can remember  
every change  
you make in a project**

# Think Git as a time machine





**vs.**



**Lives on your computer**

**Lives on the internet**

**Tracks changes**

**Stores and shares projects**

**GitHub is like the Google Drive for Git projects.**

**5 things**  **git** can do



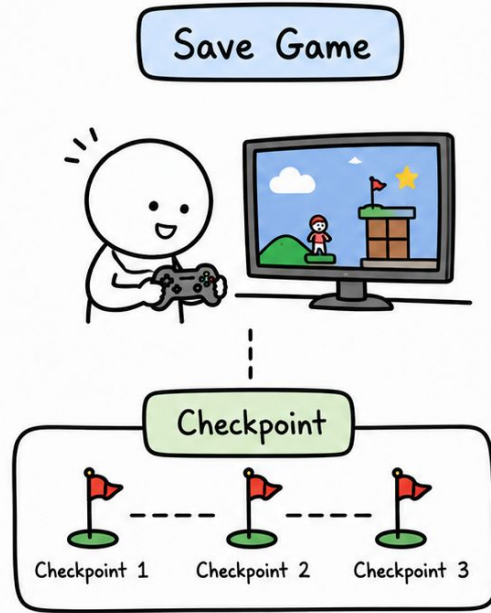
Save progress?



Save

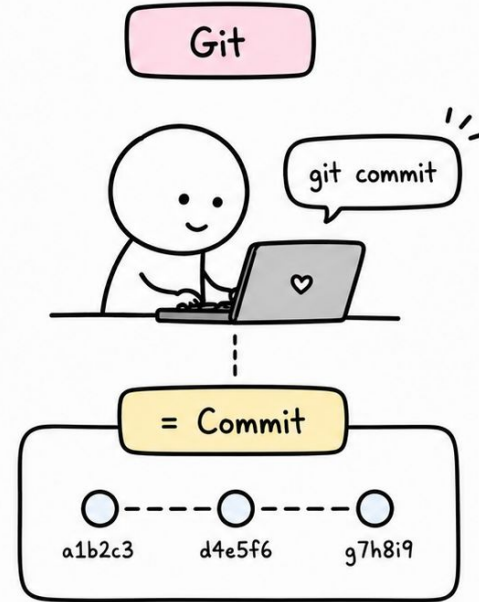
Cancel

# Thing 1: Save checkpoint



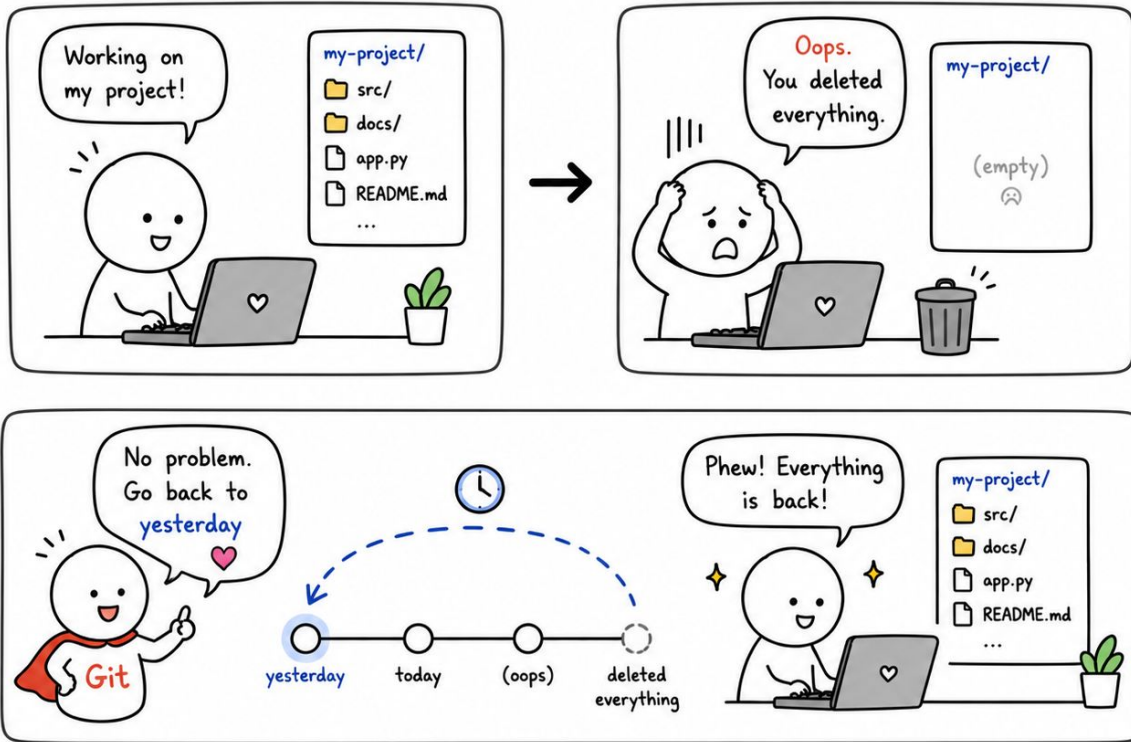
You can go back to any  
checkpoint anytime! ❤️

=

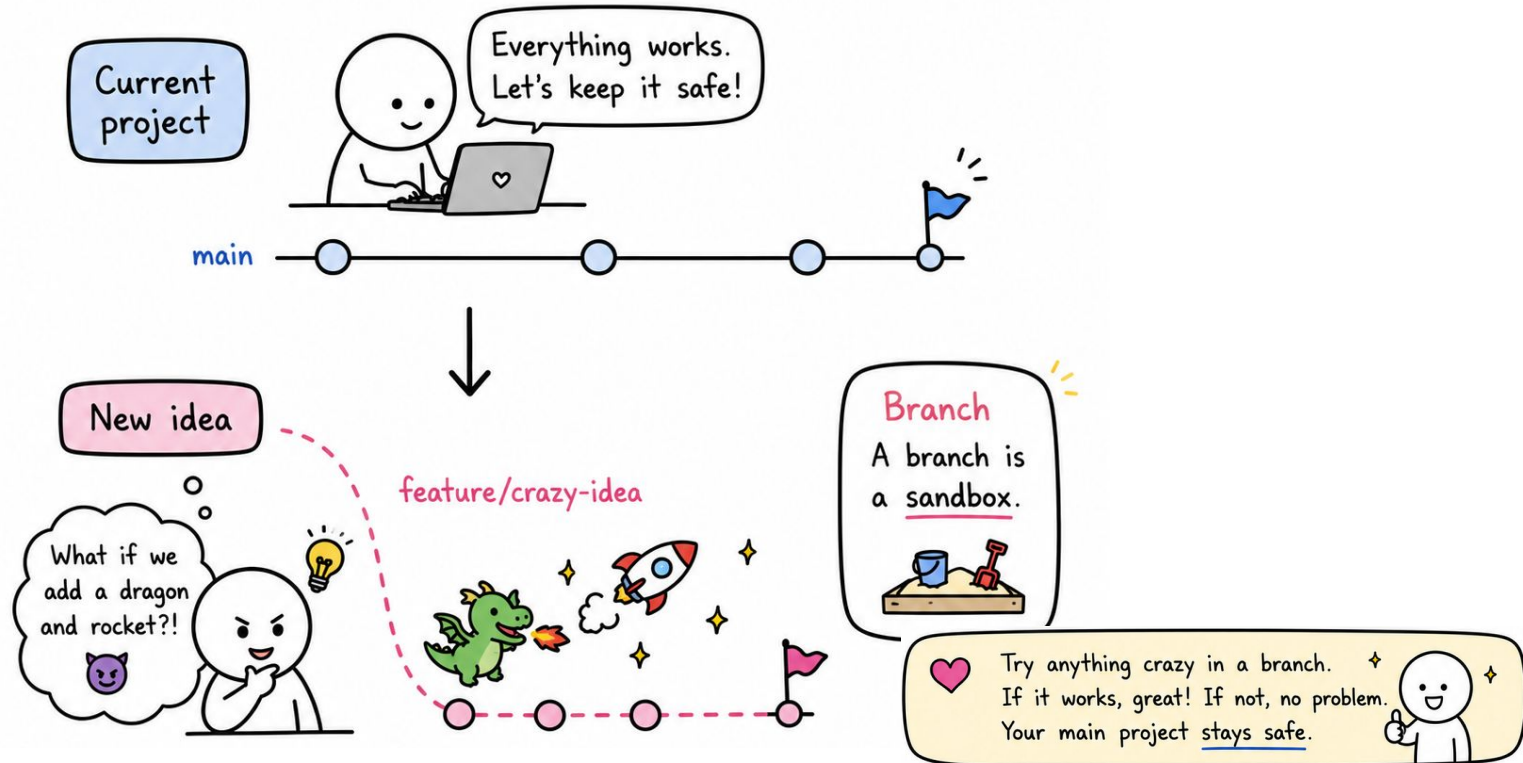


You can go back to any  
commit anytime! ❤️

# Thing 2: Time travel

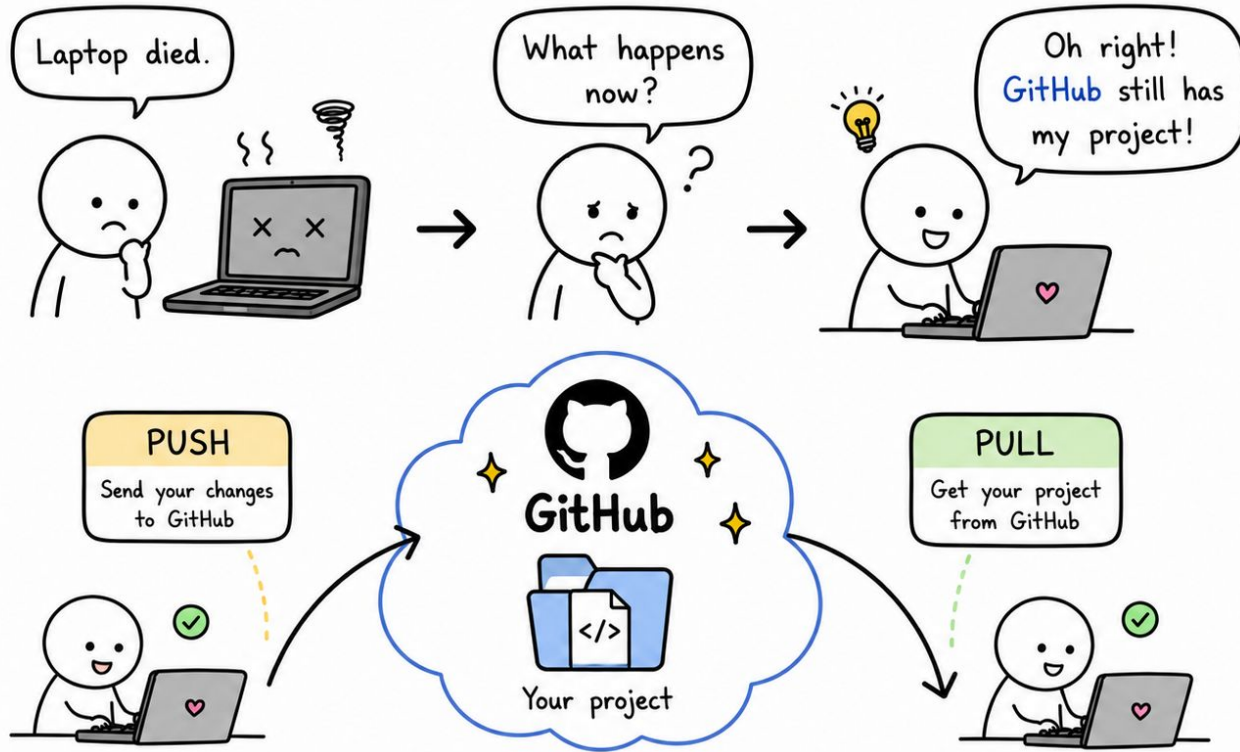


# Thing 3: Experiment safe

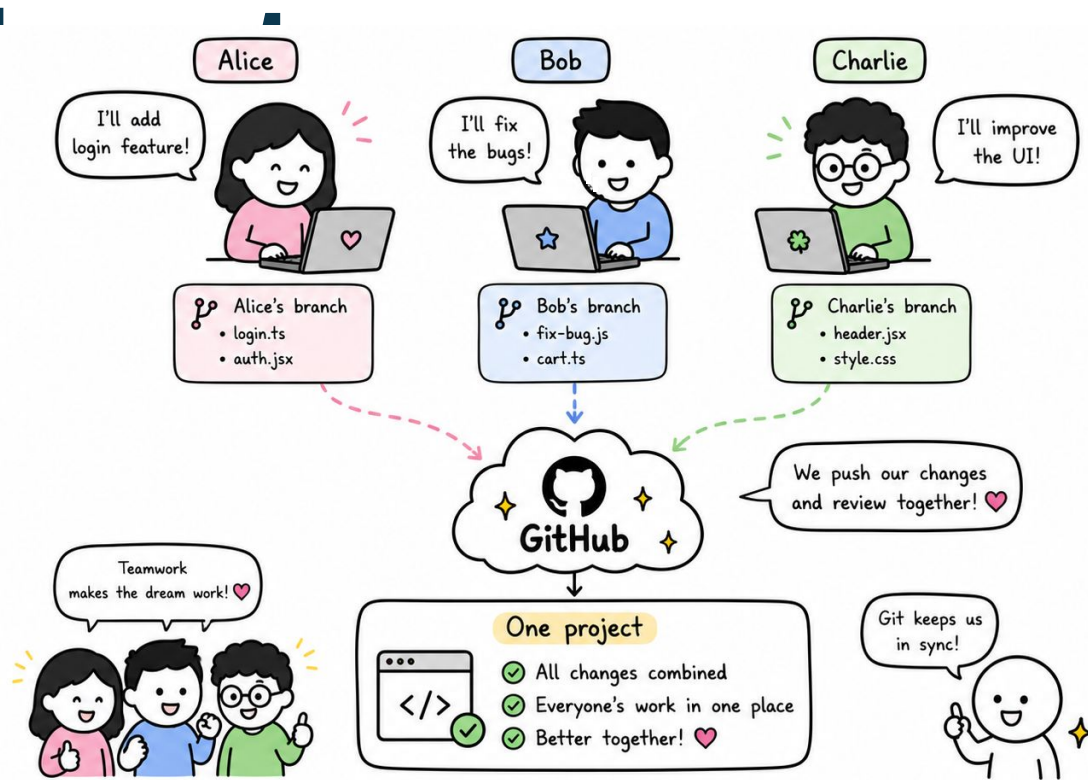


# Thing 4: Backup your

prc



# Thing 5: Help you colla'

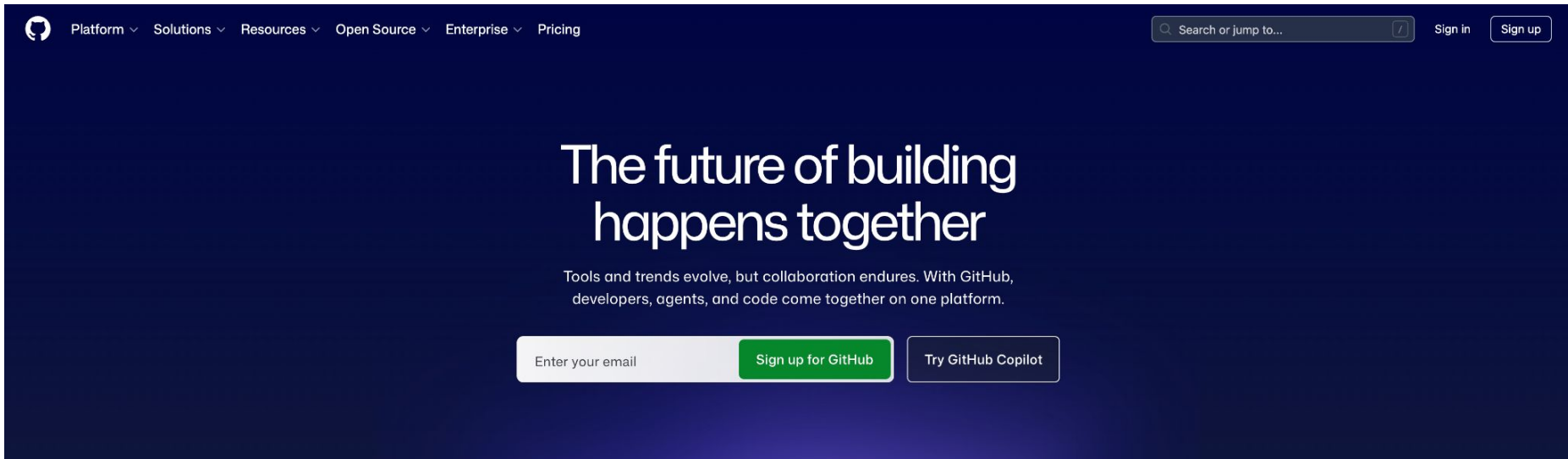


# Before we start:

| GEEK TERM   | HUMAN TRANSLATION                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| Repository  | Project Folder                  |
| Commit      | Save Checkpoint                 |
| Branch      | Safe Sandbox                    |
| Clone       | Download Project                |
| Push / Pull | Upload / Download from GitHub  |

# Step 1: Create a GitHub account and download github desktop

Google: github

A screenshot of the GitHub homepage. The header is dark blue with the GitHub logo on the left and navigation links: Platform, Solutions, Resources, Open Source, Enterprise, and Pricing. On the right of the header is a search bar with the text "Search or jump to..." and a magnifying glass icon, followed by "Sign in" and "Sign up" buttons. The main content area has a dark blue background with the text "The future of building happens together" in white. Below this is a paragraph: "Tools and trends evolve, but collaboration endures. With GitHub, developers, agents, and code come together on one platform." At the bottom, there are three buttons: "Enter your email" (white), "Sign up for GitHub" (green), and "Try GitHub Copilot" (white with a dark border).

 Platform ▾ Solutions ▾ Resources ▾ Open Source ▾ Enterprise ▾ Pricing

Search or jump to...  Sign in Sign up

## The future of building happens together

Tools and trends evolve, but collaboration endures. With GitHub, developers, agents, and code come together on one platform.

Enter your email Sign up for GitHub Try GitHub Copilot



# Create a GitHub account and download github desktop

[desktop.github.com/download/](https://desktop.github.com/download/)



**GitHub  
Desktop**

[Download](#)

[Release Notes](#)

[Help](#)

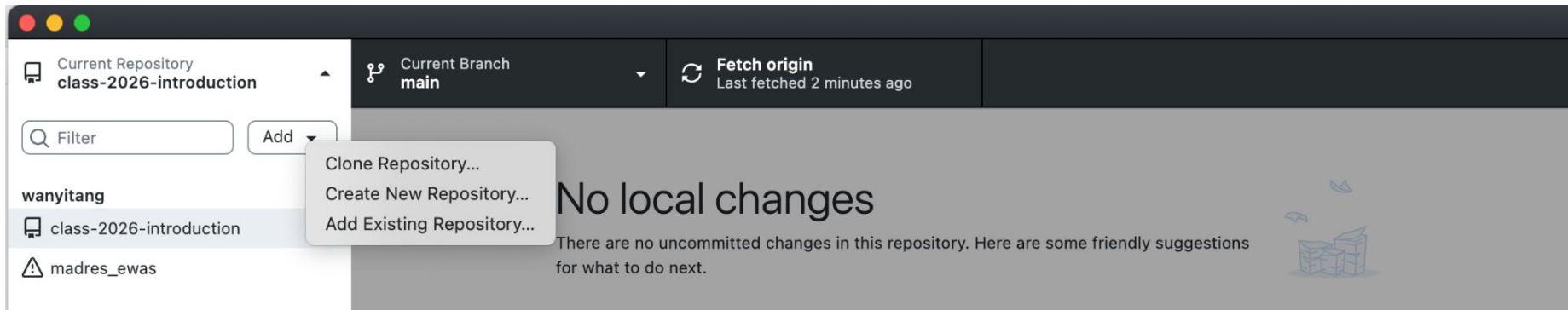
## Download GitHub Desktop for macOS

Focus on what matters instead of fighting with Git. Whether you're new to Git or a seasoned user, GitHub Desktop simplifies your development workflow.

Download for Apple silicon

# Create a repository

You can create locally using github desktop



# Create a repository

You can create locally using github desktop

NO local changes

ne friendl

## Create a New Repository

Name

TEST

Description

Local Path

/Users/wanyitang/Library/Mobile Documents/c Choose...

☒ Initialize this repository with a README

Git Ignore

None

License

None

The repository will be created at

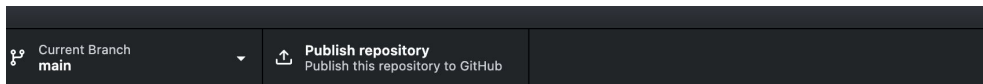
com~app n/A

TEST .

Cancel Create Repository

# Create a repository

Publish repository: Upload it to your github



## No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.



### Publish your repository to GitHub

This repository is currently only available on your local machine. By publishing it on GitHub you can share it, and collaborate with others.

Always available in the toolbar for local repositories or   P

Publish repository

### Open the repository in your external editor

Select your editor in [Settings](#)

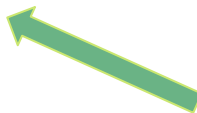
Repository menu or   A

Open in Sublime Text

### View the files of your repository in Finder

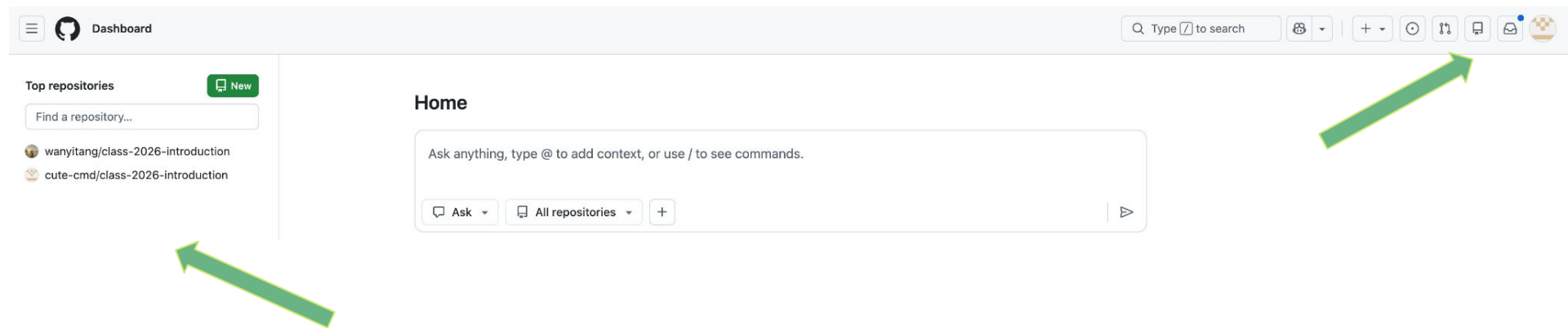
Repository menu or   F

Show in Finder



# Create a repository

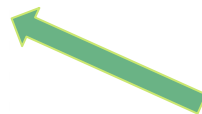
Check it in your github.com



# commit = save check point

1. Open folder and open README.md

| Name                                                                                        | Date Modified     | Size     | Kind          |
|---------------------------------------------------------------------------------------------|-------------------|----------|---------------|
|  README.md | Today at 12:22 PM | 12 bytes | Markdown File |



2. Change the README file and save it.



The screenshot shows a code editor window with a single tab titled 'README.md'. The editor contains the following text:

```
1 # TEST
2
3 ANY CHANGE YOU WANT TO MAKE
4
```

# Do your first commit

3. Check your Github page:  
NOTHING CHANGE HERE

4. COMMIT on your github desktop

The screenshot displays the GitHub Desktop application. At the top, the 'Changes' tab is active, showing a list of files to be committed: '2 changed files', '.Rhistory', and 'README.md'. Each item has a checkmark and a '+' icon. Below this, the 'README.md' file is selected, and its content is shown in a diff view. The diff shows three lines of code: a line with a minus sign and 'ROSMA PSC' (highlighted in red), a line with a plus sign and 'ROSMA P SC PROJECT' (highlighted in green), and a line with a plus sign and 'Step 1. Data exploration' (highlighted in green). At the bottom of the window, the 'Commit to main' button is visible, with a green arrow pointing to it. Above the button, there is a text input field with the placeholder 'Update README file' and a button 'Update README file'. Below the input field, there is a text area with the placeholder 'As detailed as possible'. At the bottom of the text area, there are icons for adding a commit message, a commit icon, a 'New' button, and a settings icon.

Changes 2 History

Filter

2 changed files

.Rhistory

README.md

README.md

@@ -1,2 +1,3 @@

- # ROSMA PSC

+ # ROSMA P SC PROJECT

+ Step 1. Data exploration

Update README file

As detailed as possible

Commit 2 files to main

# push = upload

5. PUSH your commit to github!

6. Check your Github page:  
NEW README!



## No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.



### Push commits to the origin remote

You have 1 local commit waiting to be pushed to GitHub.

Always available in the toolbar when there are local commits waiting to be pushed or  **P**



### Open the repository in your external editor

Select your editor in [Settings](#)

Repository menu or   **A**

Open in Sublime Text

### View the files of your repository in Finder

Repository menu or   **F**

Show in Finder

### Open the repository page on GitHub in your browser

Repository menu or   **G**

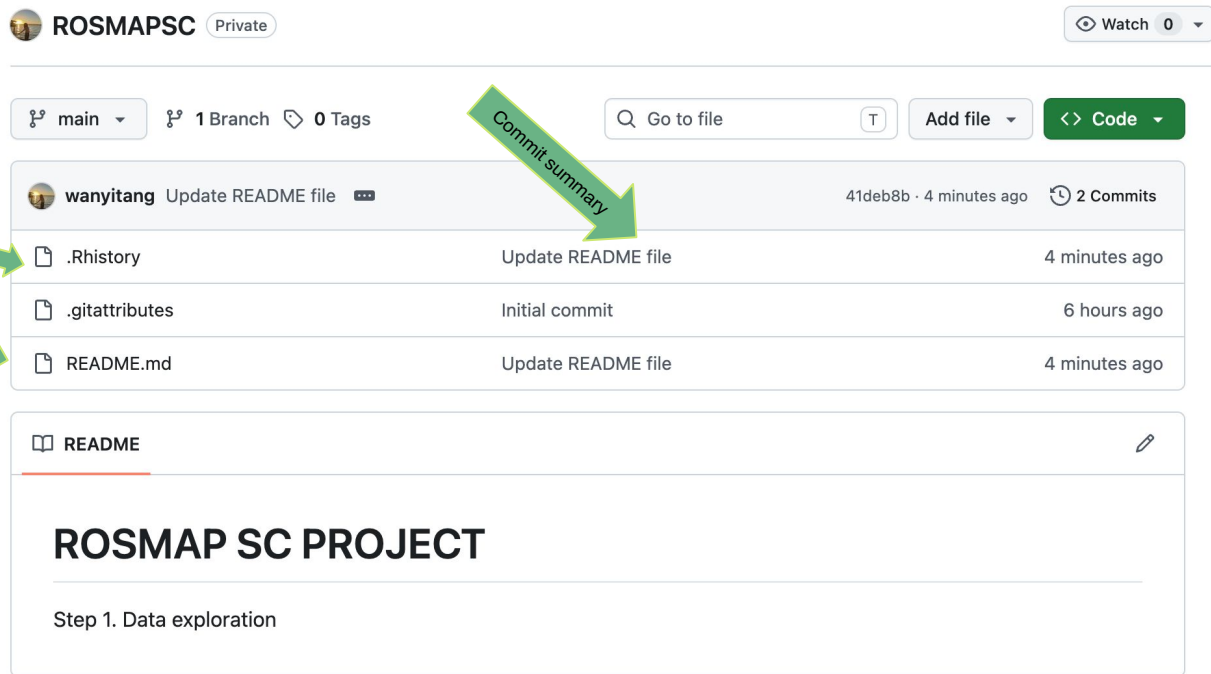
View on GitHub



# push = upload

5. PUSH your commit to github!

6. Check your Github page:  
NEW README!



ROSMAPSC Private

main 1 Branch 0 Tags

Go to file T Add file <> Code

wanyitang Update README file 41deb8b · 4 minutes ago 2 Commits

|                |                    |               |
|----------------|--------------------|---------------|
| .Rhistry       | Update README file | 4 minutes ago |
| .gitattributes | Initial commit     | 6 hours ago   |
| README.md      | Update README file | 4 minutes ago |

README

## ROSMAP SC PROJECT

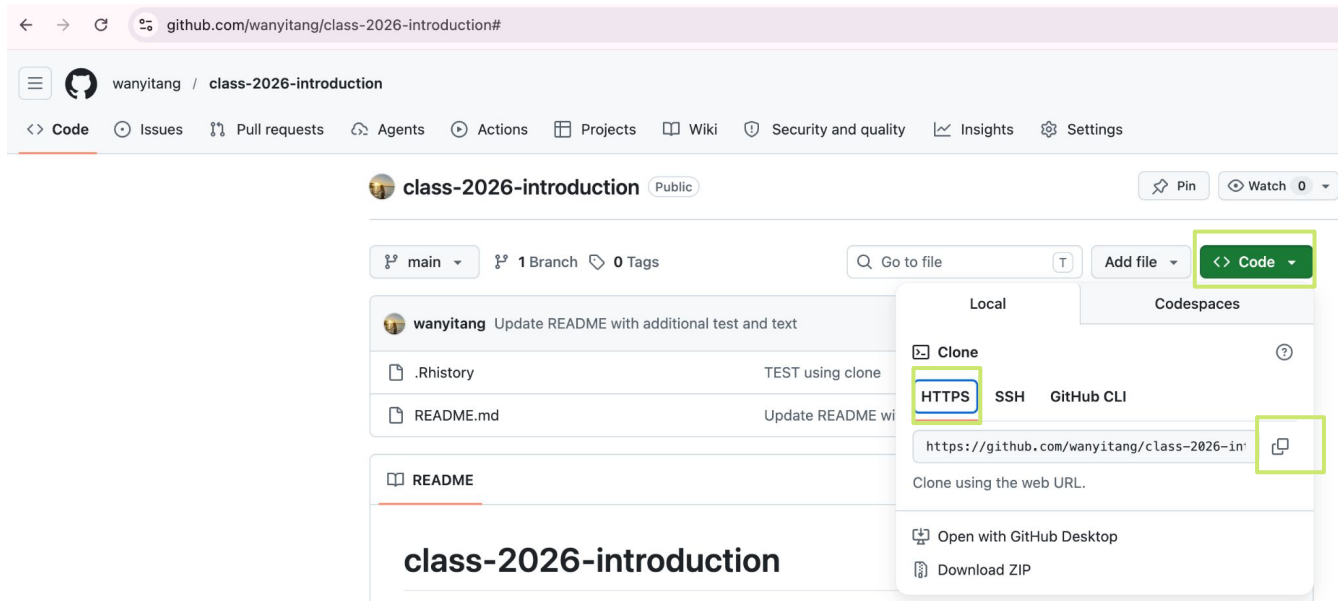
Step 1. Data exploration

# clone = save others' file

And after cloning a repository, you can use pull and push to keep your local and GitHub copies up to date!

<https://github.com/wanyitang/class-2026-introduction>

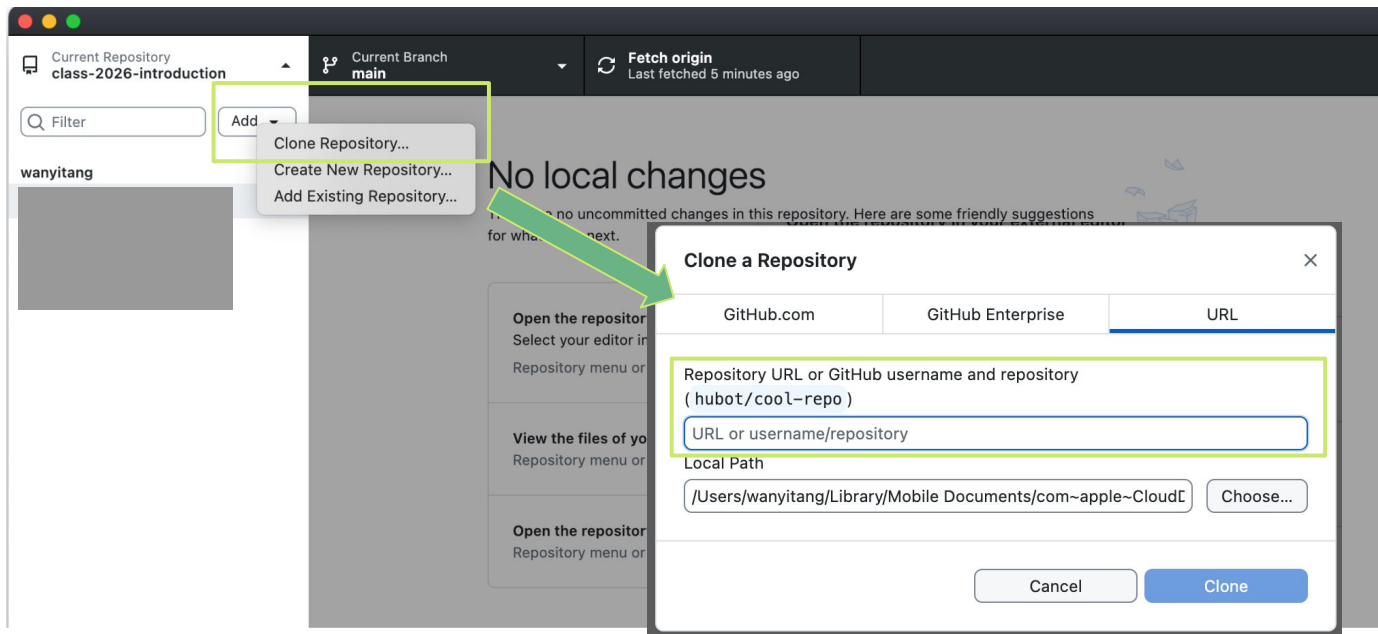
## METHOD 1



# clone = save others' file

<https://github.com/cute-cmd/firstclone>

## METHOD 1



# clone = save others' file

<https://github.com/cute-cmd/firstclone>

## METHOD 2

The screenshot shows the GitHub interface for the repository 'class-2026-introduction' by user 'wanyitang'. The repository is public and has 1 branch and 0 tags. The 'Code' button is highlighted, and the dropdown menu is open, showing options to clone the repository using HTTPS, SSH, or GitHub CLI. The 'Open with GitHub Desktop' option is highlighted with a green box. The repository name 'class-2026-introduction' is also visible at the bottom of the page.

github.com/wanyitang/class-2026-introduction#

wanyitang / class-2026-introduction

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

class-2026-introduction Public

main 1 Branch 0 Tags

Go to file Add file <> Code

Local Codespaces

Clone

HTTPS SSH GitHub CLI

https://github.com/wanyitang/class-2026-in

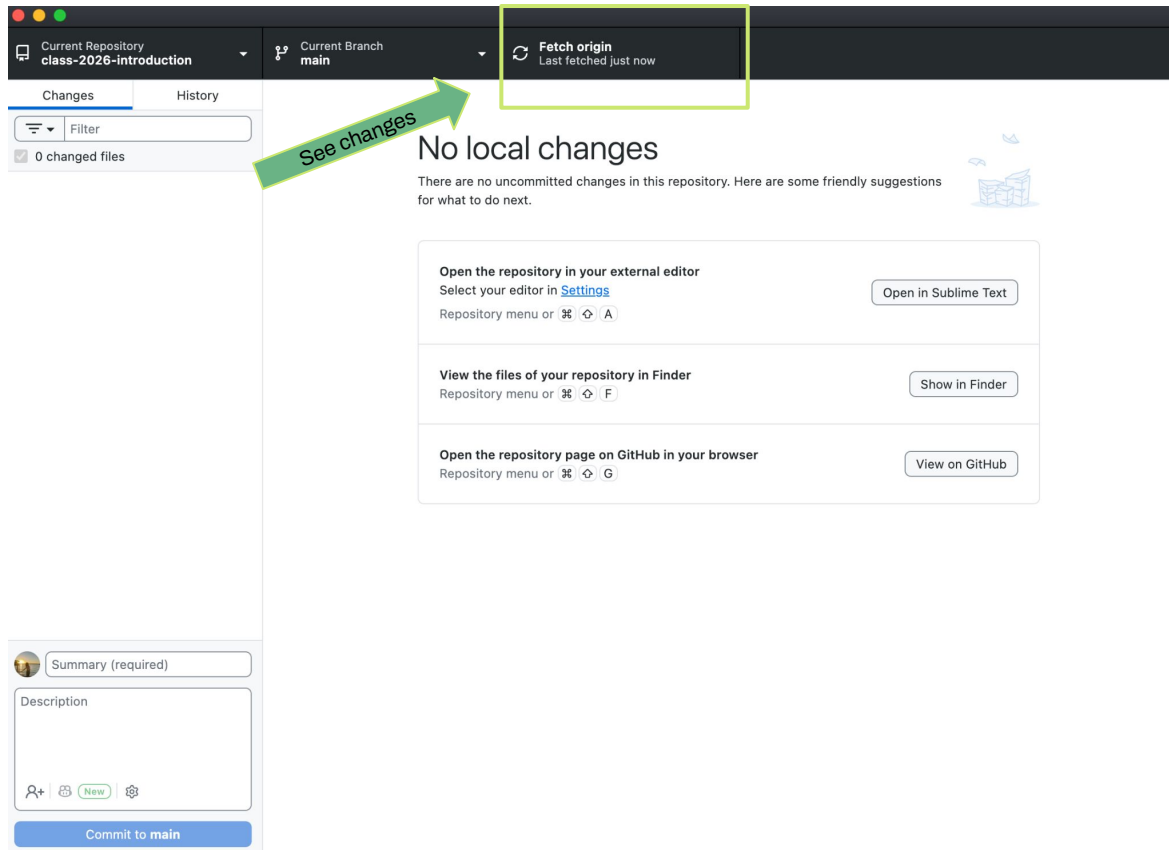
Clone using the web URL.

Open with GitHub Desktop

Download ZIP

class-2026-introduction

# pull = download



The screenshot shows the GitHub Desktop application interface. At the top, the 'Current Repository' is 'class-2026-introduction', the 'Current Branch' is 'main', and the 'Fetch origin' button is highlighted with a green box. A green arrow points to the 'Fetch origin' button with the text 'See changes'. Below the top bar, the 'Changes' tab is selected, showing '0 changed files'. The main area displays 'No local changes' and a message: 'There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.' Below this, there are three suggestions:

- Open the repository in your external editor**  
Select your editor in [Settings](#)  
Repository menu or    A  
[Open in Sublime Text](#)
- View the files of your repository in Finder**  
Repository menu or    F  
[Show in Finder](#)
- Open the repository page on GitHub in your browser**  
Repository menu or    G  
[View on GitHub](#)

At the bottom, there is a 'Commit to main' button and a 'Summary (required)' field. The 'Description' field is also visible.

# pull = download

Current Repository  
class-2026-introduction

Current Branch  
main

Pull origin  
Last fetched just ...

Changes History

Filter

0 changed files

## No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

**Pull 1 commit from the origin remote**  
The current branch ( main ) has a commit on GitHub that does not exist on your machine.  
Always available in the toolbar when there are remote changes or ⌘ ⇧ P

**Pull origin**

**Open the repository in your external editor**  
Select your editor in [Settings](#)  
Repository menu or ⌘ ⇧ A

**Open in Sublime Text**

**View the files of your repository in Finder**  
Repository menu or ⌘ ⇧ F

**Show in Finder**

**Open the repository page on GitHub in your browser**  
Repository menu or ⌘ ⇧ G

**View on GitHub**



**is not an app**

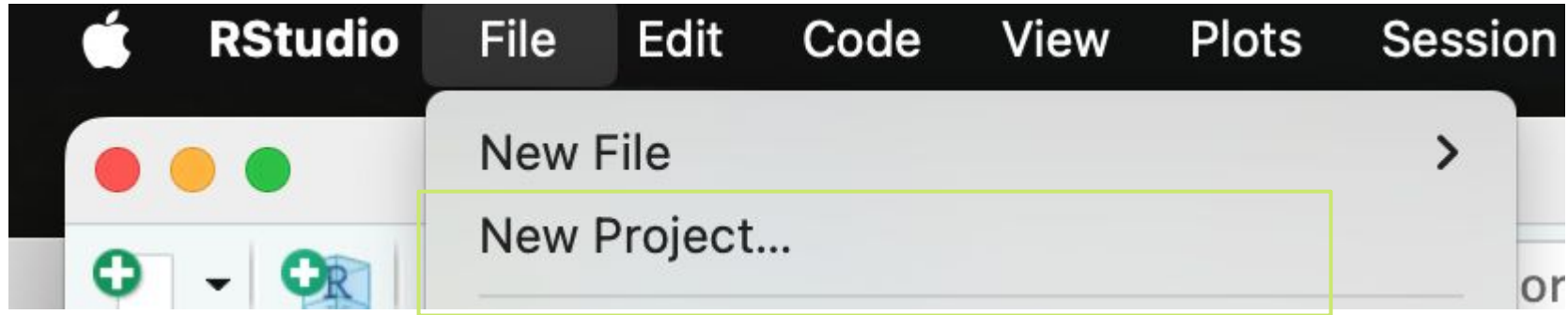
**is a version control tool**

You can use it on lots of platforms

# Use git with R studio

<https://github.com/USCbiostats/LAsBEST>

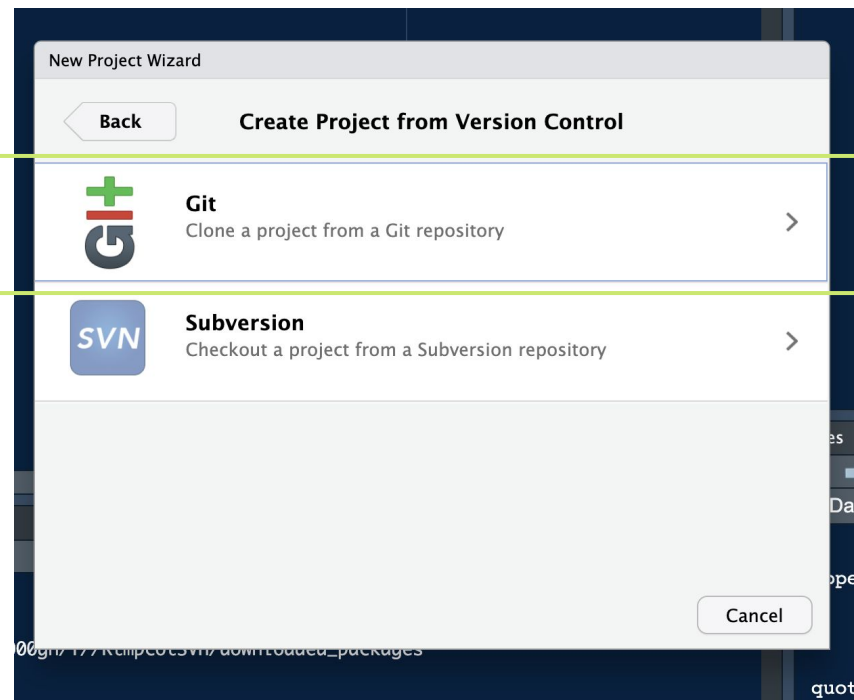
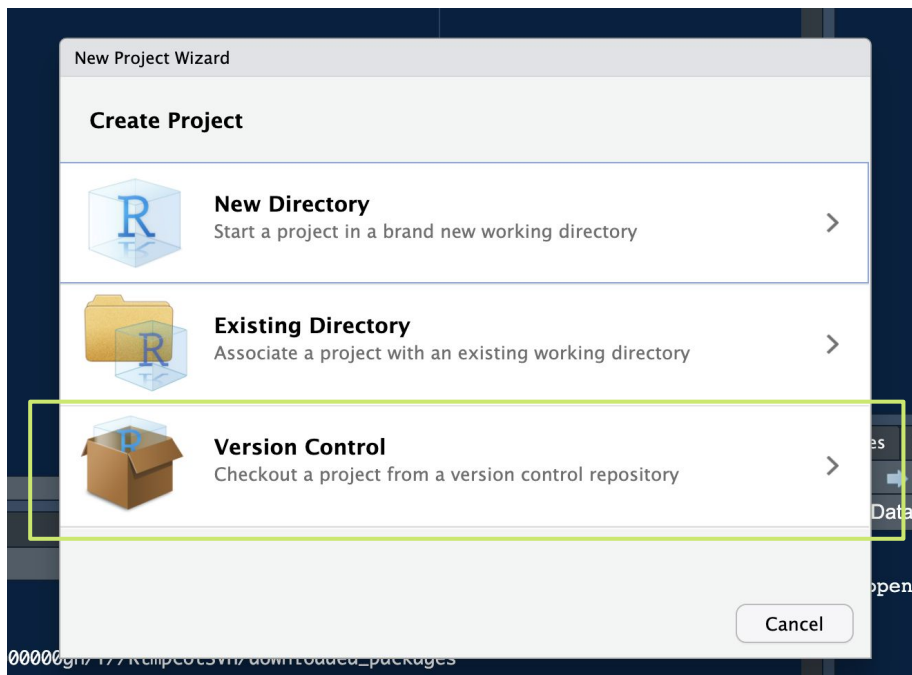
Step 1





# clone with R studio

<https://github.com/USCbiostats/LAsBEST>



# clone with R studio

<https://github.com/USCbiostats/LAsBEST>

New Project Wizard

Back Clone Git Repository



Repository URL:

Project directory name:  
 Folder name

Create project as subdirectory of:  
 Path

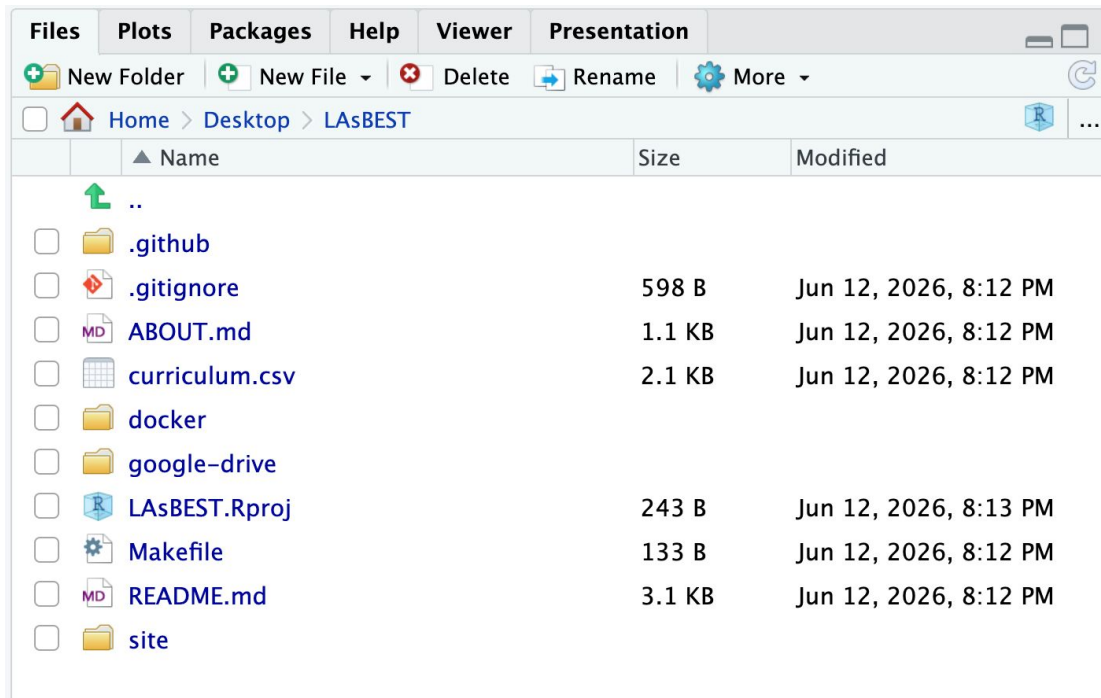
☐ Open in new session

- Clone the USCbiostats/LAsBEST repository from GitHub.
- Create a local project folder at **~/YOUR\_PATH/LAsBEST**.
- Open the project in RStudio.
- Verify that the project name LAsBEST appears in the upper-right corner of the RStudio window.

# clone with R studio

<https://github.com/USCbiostats/LAsBEST>

Check the R files in R studio!



- Clone the USCbiostats/LAsBEST repository from GitHub.
- Create a local project folder at `~/YOUR_PATH/LAsBEST`.
- Open the project in RStudio.
- Verify that the project name LAsBEST appears in the upper-right corner of the RStudio window.

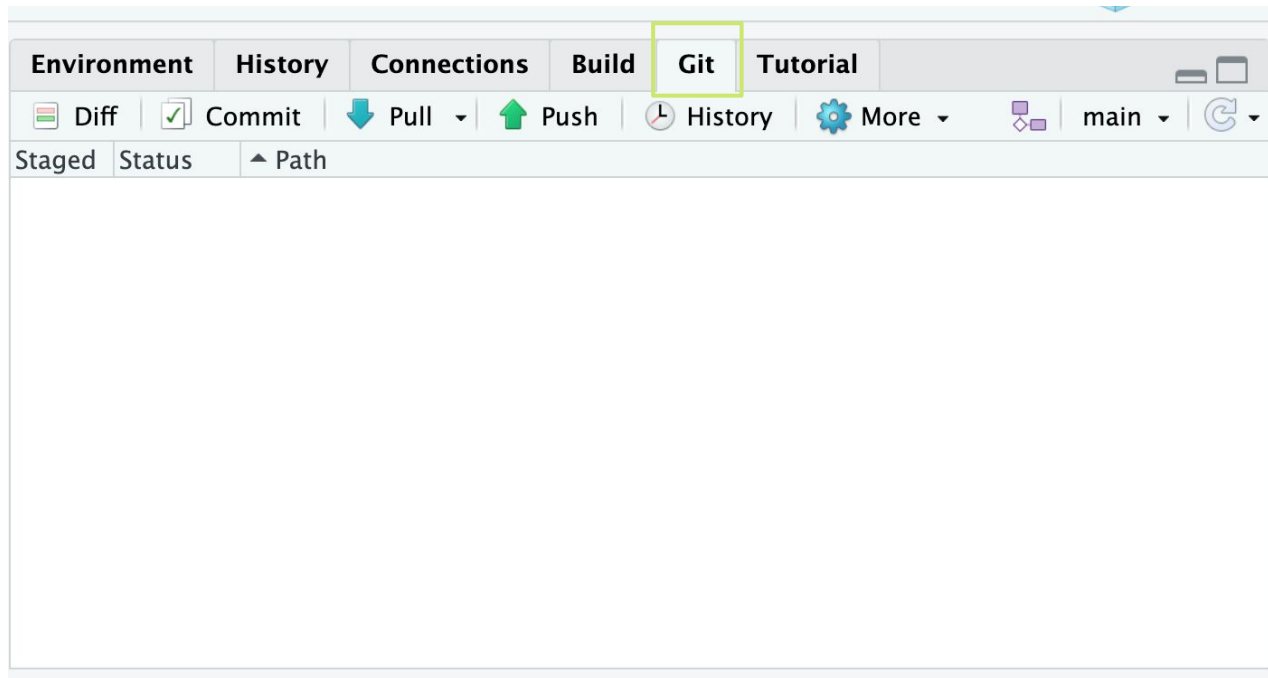
# clone with R studio

Check the folder!

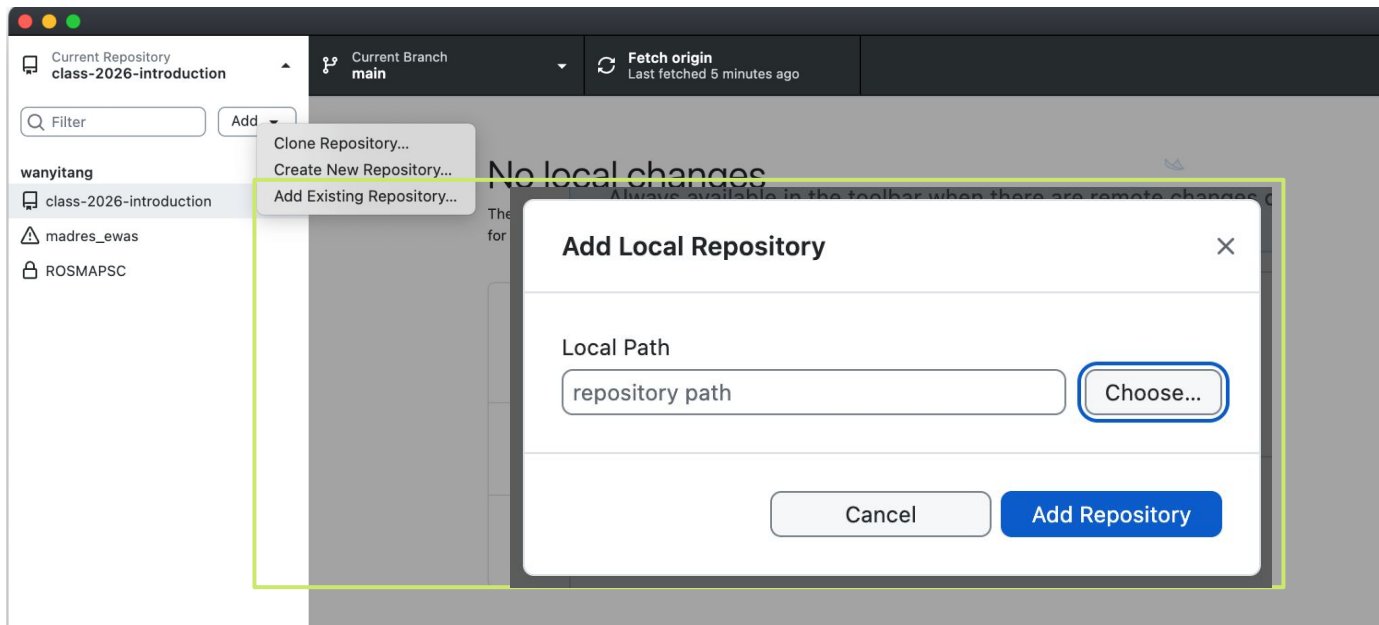
<https://github.com/USCbiostats/LAsBEST>

| LASBEST                                                                                          |                  |           |                  |
|--------------------------------------------------------------------------------------------------|------------------|-----------|------------------|
| Name                                                                                             | Date Modified    | Size      | Kind             |
|  ABOUT.md       | Today at 8:12 PM | 1 KB      | Markdown File    |
|  curriculum.csv | Today at 8:12 PM | 2 KB      | Comm...et (.csv) |
| >  docker       | Today at 8:12 PM | --        | Folder           |
| >  google-drive | Today at 8:12 PM | --        | Folder           |
|  LASBEST.Rproj  | Today at 8:13 PM | 243 bytes | R Project        |
|  Makefile       | Today at 8:12 PM | 133 bytes | Plain Text       |
|  README.md      | Today at 8:12 PM | 3 KB      | Markdown File    |
| >  site        | Today at 8:12 PM | --        | Folder           |

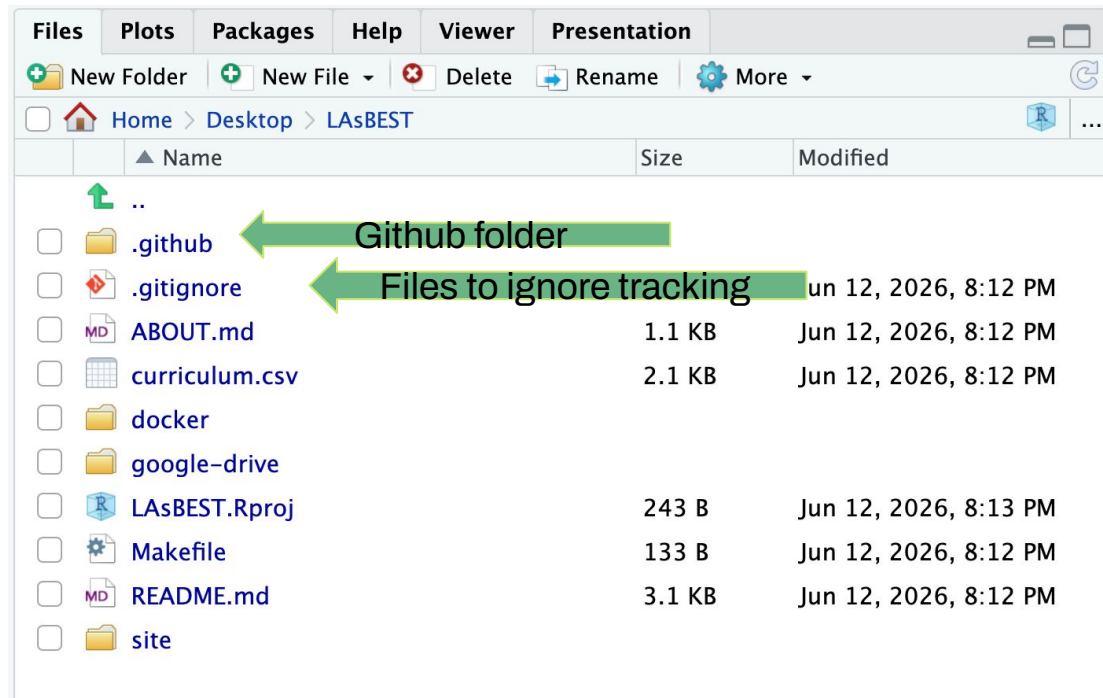
# Use git with R studio



# IF you prefer using Github Desktop instead of Rstudio to git



# How to read a github repository



ALWAYS starts with README!

**\*Try collaboration  
using **FORK** if have  
time**



# Resources:

- Book Happy Git and Github for the useR:
- <https://happygitwithr.com/big-picture>
- Video tutorial to set-up Git and Github in Rstudio:
- <https://rfortherestofus.com/2021/02/how-to-use-git-github-with-r>
- Youtube video explaining Git basics:
- <https://youtu.be/eWxxfttcMts>
- Paper: Excuse me, do you have a moment to talk about version control?
- <https://peerj.com/preprints/3159v2/#>